

LESSON 2-2 • SESSION 1 • PRACTICE • FORM A**Building a Histogram**

Learning target – I can group data into equal intervals and build a histogram from a frequency table. • 6.SP.B.4

REMEMBER

- An **interval** is a group of numbers, like 10–19. Intervals are equal in width and **never overlap**.
- **Frequency** = how many values fall in an interval. It sets the **bar height**.
- Histogram bars **touch**. All the bar heights add up to the number of values.

WORKED EXAMPLE

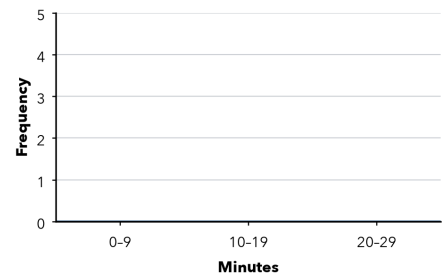
Walk times (min): 4, 15, 9, 22, 11, 18, 27. Use 0–9, 10–19, 20–29.

Step 1 0–9 holds 4 and 9: frequency 2.**Step 2** 10–19 holds 15, 11, 18: frequency 3. 20–29 holds 22, 27: frequency 2.**Step 3** Draw touching bars of height 2, 3, 2.**Answer Check:** $2 + 3 + 2 = 7$ values**WORDS** histogram (*histograma*) • interval (*intervalo*) • frequency (*frecuencia*)**PRACTICE***show your work • label units*

- 1. Bus Rides.** Nine students timed their bus rides (min): 12, 25, 8, 17, 21, 14, 6, 23, 19. Complete the table. Then draw the histogram.

Hint Cross off each value as you tally it.

Interval	Frequency
0–9	
10–19	
20–29	

**Most rides** _____

- 2. Pushups.** 8 students did these pushups: 22, 15, 9, 27, 13, 18, 26, 24. Find each frequency and the total.

a. 0–9: ____ **b.** 10–19: ____ **c.** 20–29: ____ **d.** Total: ____

- 3. Read the Table.** 40–49: 5, 50–59: 8, 60–69: 4, 70–79: 3.

Hint Add the frequencies; the biggest one is the tallest bar.**a.** Values in all: ____**b.** Tallest bar: ____

- 4. Missing Bar.** A histogram shows 9 values. Two bars have heights 2 and 4. How tall is the third bar?

Hint All the bar heights add up to the total.

Show your work.

Height _____

- 5. Car Wash.** Cars washed on 7 Saturdays: 12, 27, 18, 33, 24, 9, 30. Find each frequency.

Hint Put the counts in order first.

a. 0–9: ____ **b.** 10–19: ____ **c.** 20–29: ____ **d.** 30–39: ____

- 6. Overlapping Bins.** Lena sorts 5, 11, 16, 20, 24 into the intervals 5–10, 10–15, 15–20, 20–25. What is wrong? Write intervals that work.

Hint Look for a number that fits in two intervals.

- 7. Choose the Intervals.** Test scores run from 51 to 98. Write five equal intervals of width 10 that cover every score.

Intervals _____

- 8. Field Day.** Twelve runners finished in 31, 44, 38, 52, 47, 35, 41, 56, 39, 48, 33, 55 seconds. Complete the table.

a. Which interval holds the most runners? _____

b. Why does a histogram show the times better than the list?

Interval	Frequency
30–39	
40–49	
50–59	

FIND THE ERROR

explain the mistake, then fix it

- 9. Ibrahim's Work.** For the values 4, 12, 18, 26, Ibrahim drew bars over 0–9, 10–19 and 20–29 with a space between each bar. Explain the mistake and describe the correct histogram.

TEST PRACTICE

choose the best answer

- 10.** In a histogram, the height of a bar shows the –
- A** width of the interval **B** frequency of the interval
- C** greatest value **D** number of intervals
- 11.** A frequency table shows 3, 7, 5 and 2. How many values were collected?
- A** 7 **B** 4 **C** 17 **D** 15

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

8 L	6 Z	50–59 I	10–19 T	30–39 S	4 A	3 E	20 L	20–29 X	9 Q
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#1	#2c	#2d	#3a	#3b	#4	#8a
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CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #____
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